MOHAMED AMER

Data Scientist & Machine Learning Engineer

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Professional Summary

Data Scientist and Machine Learning Engineer with hands-on experience developing predictive models and optimizing complex processes at Volvo SML. Proficient in Python, TensorFlow, and PyTorch, with a strong background in leveraging AI to solve real-world logistical and industrial challenges.

Education

Halmstad University — **Halmstad, Sweden**
M.Sc. Artificial Intelligence — *Expected June 2026*

Halmstad University — **Halmstad, Sweden**
B.Sc. Computer Science — *June 2023*

Work Experience

Student Researcher — **Nov. 2024 – June 2025**
Volvo SML — *Gothenburg, Sweden*

- Integrated into the on-site logistics team at Volvo SML to analyze and streamline complex supply chain processes.
- Developed and evaluated predictive machine learning models to optimize logistics flows across large-scale datasets.
- Collaborated directly with field engineers to translate data-driven insights into deployable efficiency improvements.

Research Publications

Prediction Intervals for ML-driven Logistics | *Master's Thesis at Volvo SML* — **Nov. 2024 – June 2025**

- Developed demand forecasting models with prediction intervals to improve inventory management.
- Implemented discrete-event simulation and Beta-PERT distributions, reducing costs while maintaining high service levels.

Room Booking and Access Control System | *Bachelor's Thesis at Halmstad University* — **Jan. 2023 – June 2023**

- Built an integrated system for room bookings and access control using NodeJS, Android Studio, and PostgreSQL.
- Implemented secure Mobile BankID authentication and developed mobile apps to resolve booking conflicts and unauthorized access.

Other Projects

Chest X-Ray Classification

- Developed a CNN model using TensorFlow for medical image analysis.
- Achieved 92% validation accuracy.

Spare-Parts Demand Forecasting

- Built a complete MLOps pipeline using Prophet, LightGBM, and FastAPI.
- Implemented end-to-end machine learning workflow for demand prediction.

Pattern Finder – Trading Patterns with Computer Vision

- Developed a system using computer vision to identify technical patterns in trading charts.
- Implemented matching against historical crypto data to find similar price scenarios.

Technical Skills

Languages: Python, Java, JavaScript, C, SQL, MATLAB
Machine Learning & Data: TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy
Generative AI & LLMs: RAG, Prompt Engineering, Claude Code, GenAI Architectures
MLOps & Infrastructure: CI/CD, Docker, Git, REST APIs, Cloud Platforms
Core Concepts: Advanced OOP, Data Structures & Algorithms, Statistical Modeling

Languages & References

Languages: Swedish (Native), English (Advanced C1), Arabic (Native)
References: Available upon request.